

Developing an EDA Framework for Electronic Textiles

Case Study: Characterization & Design Rules Development for Stitched E-Textiles

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ABSTRACT

Electronic textiles (e-textiles) development and manufacturing processes have largely been based on individual (Do it yourself) DIY manufacturing procedures without rigorous production process standardization. Because of this, textile circuits have mostly been subjected to a trial & error development approach. In line with this motivation, this dissertation will investigate, identify, and synthesize e-textile fabrication parameters to develop a design guideline for woven stitched textile circuits. The first phase investigates and synthesizes expert knowledge that will allow for the formulation of design rules regarding the manufacturability of an e-textile. The second phase involves the development of design rules for e-textiles manufacturability to standardize the development process of e-textiles in a more reliable and repeatable way, especially because repeatability and reliability are the most important factors in manufacturability [1]. The third phase involves the deployment of these rules in a solar e-textile circuit application.

CCS CONCEPTS

• **Electronic design automation**; • **Printed circuit boards**; • **Human computer interaction (HCI)**;

KEYWORDS

Design rules, Electronic textile circuits, Characterization & standardization, Smart textiles, Standards

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1 PROBLEM STATEMENT

Printed Circuit Boards (PCB) are well-established in traditional electronics. However, they do not translate well to e-textiles and wearable systems, which require more flexibility than traditional PCBs

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[2],[3]. Most conventional textile manufacturing techniques were not designed for rigid components, and vice versa. [19] mentioned a need for optimizing components joining to e-textile applications, as most were built or developed to interface with traditional PCB and not e-textile circuits. For instance, modeling the relationship between pad geometry, lead geometry, and soldering provides insight for improvement by the integrated circuit industry for e-textile circuit applications. Navigating the complexity of a soft, flexible fabric substrate for electronics is non-trivial but essential to making textile circuit applications a reality for mass production [4]-[6]. Compared to their PCB counterparts, there is no standardization in the production of e-textiles circuits. E-textile circuits are mostly developed through trial-and-error methods, whereas PCBs are produced worldwide with precise production stages and processes [7],[8]. In conventional PCB manufacturing, pads are usually 1.8 times larger than the hole size for through-hole technology and 1.2 - 1.5 times larger than component lead sizes for surface mount technology. Also, it is recommended in PCB that components are soldered in the middle of pads to improve stability and durability. E-textile circuits do not have any such rules to guide our manufacturing process. This production disparity is understandable because e-textile circuits are relatively new and primarily in developmental stages. As a starting point for standardizing the e-textile circuit manufacturing process, this dissertation aims to investigate and synthesize e-textile circuit manufacturing parameters and their relationships from expert knowledge. Expert knowledge will then be formalized into an Electronic Design Automation framework, and characterization studies will be used to inform design rules to help guide the development and manufacturing of e-textile circuits. The case study of stitched manufacturing processes on woven fabric substrates is explored in depth.

2 RELATED WORK

2.1 E-textile Standards

The major reason for the absence of standards for e-textile is that it is a relatively new field in the world of electronics and because of the many criteria and sub-criteria that need to be considered in an e-textile system (from the type of fabric to the type of conductive yarn to be used) based on individual applications. While trial-and-error methods can be feasible for DIYs and small-scale projects, standardization is needed if e-textile manufacturing processes are to be commercialized. The Interconnecting and Packaging and Components (IPC) organization has started developing some new standards for woven, knitted, printed, and stitched e-textiles. There are six major categories of standards IPC is working on: requirements for conductive yarns for e-textile applications (IPC-8911), requirements for woven/knitted/braided e-textiles integrated with conductive yarns/wires (IPC-8921), qualification and performance

specification for printed electronics on coated/treated e-textiles (IPC-8922), design standards for embroidered e-textiles (IPC-8953), quality and reliability of e-textile wearables (IPC-8981), and guidelines on e-textiles wearables (IPC-8961). The IPC standards are the first baseline compilation of definitions and terminologies from the textiles and electronics industries in one document [9]. The IPC-8971 and IPC-8952 standards, which are variations of the IPC-8922, are some of the released standards. They establish the test requirements for printed e-textiles on a coated textile substrate and their components. These standards have established the layout design and electrical & mechanical tests essential for printed e-textiles. For example, in the IPC-8952 standard, it was specified that the minimum width and thickness of conductors on printed e-textiles are based on the current capacity and maximum permissible temperature. There is still a need to understand how these recommendations apply to other fabrication methods and to translate guidelines into specific, measurable values and relationships in order to develop design rules to support design automation.

2.2 Electronic Design Automation

Electronic Design Automation (EDA) is a crucial and vital part of modern electronic design (e-textile design inclusive). EDA involves tools and software algorithms to automate various design steps of the electronic design process. Design rules can significantly improve the manufacturability and performance of e-textile circuit design, eliminating some of the trial-and-error steps in current processes. Design rules optimization through technology has optimized automatic circuit layouts in traditional PCBs. This optimization has led to the development of auto-routing in PCBs [10]. There is presently no EDA tool in e-textile circuit manufacturing because of the absence of characterized variables in e-textile fabrication methods. The closest we have to developing EDA rules for e-textile circuits was a study conducted by Locher et al. [11]. That study adapted EDA tools for printed wiring boards (PCB) to design auto-routing and component placement rules in woven e-textiles [11]. But the focus was more on using regular PCB design tools than understanding how to adapt them to e-textile manufacturing. E-textiles can be manufactured via weaving, knitting, stitching, printing, etc.; although each method has different variables and parameters, some aspects will likely transcend fabrication methods. Here, the stitching method is used as a case study while exploring the translation of the EDA framework to other fabrication methods. Some examples of the variables/parameters that will need to be characterized and formalized into EDA rules for the stitching fabrication method include the following, among many others: stitch length, stitch density, stitch type, stitch spacing, fabric thickness, conductive thread types, conductivity, durability, etc. [12]–[17]. Our study will focus on the stitching fabrication method as a case study to identify, characterize and synthesize variables and relationships into EDA rules needed to build an EDA tool for e-textiles, not developing software algorithms for design automation. Further, while EDA tools encompass a wide variety of simulation and prediction functionality, here, the focus is on design rules related to layout and routing.

2.3 Previous Work

In our previous work [18], we created a solar e-textile by integrating small beaded solar cells using stitched fabrication methods. We then characterized the influence of some fabrication parameters on energy output. Characterizing the manufacturing process of this garment led to experiences and insights that motivated the need for an EDA framework for e-textiles. For example, the characterization of the minimum stitch spacing for routing traces on the shirt led to an understanding of the importance of accounting for fray diameter in stitched, uninsulated conductors. Frays are unraveling individual strands of threads from the main body of the thread, and they introduce a risk of short circuits. Hence, the typical thread diameter (similar to a typical trace width measurement) alone is insufficient for e-textile circuit design, and understanding the maximum fray diameter possible is necessary. In our case, a double-stitched Vectran thread was about 1.09mm, while a frayed Vectran stitch was about 4mm in diameter. Another major challenge we observed from this work was the straightness of the stitch, even for CNC-controlled automatic stitching machines. Straightness of stitched traces can depend on many factors like the needle position, bobbin tension, the tension between the needle and fabric, alignment of fabric grain with the stitch line, etc. Hence, the stitched trace layout is not as precise as in traditional CAD PCB design software and must be accounted for in layout and routing. Many other factors still have not yet been standardized for e-textile circuit design, and a few of them have been highlighted in Table 1.

3 METHODOLOGY & EVALUATION

3.1 Phase 1 – Descriptive Research

This study will be conducted in three major phases. Phase 1 focuses on developing the variables and relationships framework for e-textile circuit design rules (related to routing and placement) from expert knowledge. A qualitative research method will be used to identify common variables and relationships. In phase 2, we will implement and evaluate (and improve if need be) a smaller number of selected variables to characterize these relationships and variables. Finally, in phase 3, these rules will be used to develop a solar e-textile circuit. The first phase of the study focuses on identifying the layout parameters (independent variables), e.g., stitch length, stitch spacing, stitch type, etc. and performance metrics (dependent variables), e.g., functionality, durability, washability, etc. that describe and characterize e-textile circuit's routing from previous literature, experts' interviews, and researcher's experience with e-textiles.

3.1.1 Data Collection Method(s). We will have three major sources of data collection methods for this study: (i) exhaustive search for literature/literature scraping/compilation of variables in pieces of literature, (ii) in-depth semi-structured interviews with experts, and (iii) researcher's insights and experience. The exhaustive literature search will compile literature from the year 2000 till date. This approach is because 2000 was the year e-textile wearable technology became common [19]. The second source of data collection method is semi-structured interviews with e-textile experts to share their experiences, expertise, and personal perspectives about their interaction with the design and fabrication of e-textiles. We will

Table 1: The major distinctions/similarities between a printed circuit board and an e-textile circuit.

Printed Circuit Board	E-textile Circuit
Known materials for etching (e.g., Copper)	No specific conductive thread (depends on the application)
Copper traces thickness is 0.035 mm.	None
Circuit board thickness = 0.8, 1.6, or 2.4mm	Fabric thickness is usually unspecified.
Universal measurement unit	None
Boards laid out on a fixed grid – snap grid.	No fixed grid. It depends on where the stitch is routed from.
Minimum spacing between 240volts tracks and isolated signal tracks – 8mm.	None
Every track has its resistance.	Not necessarily the same for conductive threads. Sometimes, resistance depends on the material the thread is made of.
Narrower track = higher resistance.	None
Pad-to-hole ratio – minimum of (1.8x hole or 0.5mm larger)	None (even though useful for a multilayer circuit)
Vias to hole ratio = 0.5 – 0.7mm	None
Clearance for pads and holes = 15 thou or 8-10 for dense layouts	None
There are equations and charts that guide the electrical properties of each track. The general rule of thumb is that thicker tracks are preferable.	None
Usually on a rectangular shape board and usually smaller sizes	Irregular shapes and larger sizes across the body silhouette
Short routing is encouraged for both e-textile circuit and PCB. It decreases resistance, capacitance, and inductance.	None
Recommended angle for routing = 45°. Avoid 90° or more.	Getting precise rounded corners in e-textile circuit can be tricky.
Avoiding sharp corners is recommended for PCB.	Serpentine tracks are not yet perfected.
Perfected serpentine track corners	Through-hole component legs to connect top and bottom tracks
The substrate is usually Fiberglass with known material properties. The dielectric constant value of 3.9-4.8 (Sometimes, Teflon is used)	No standardized fabric material properties are known for a textile circuit substrate.
Netlists (in Gerber/drill/pick and place file formats)	The closest we have are layers of dxf/pdf files in AI/CLO3D

recruit a minimum of five and a maximum of ten experts, and each should have a minimum of one year's experience in interaction with e-textiles with no age limit restriction. A semi-structured interview format will be used to understand the participant's deeper insights and perspectives on e-textile circuits. The third source of data collection is the researcher's experience. The researcher has experience interacting with e-textiles, and the researcher's perspective and insights will be an additional source of data in developing stitched e-textiles design rules. Multiple data collection methods (also called triangulation) reflect an attempt to secure an in-depth understanding of a phenomenon in a qualitative study [20]. The triangulation processes add rigor, richness, and breadth of complexity to the study.

3.1.2 Data Analysis. An inductive qualitative grounded theory approach using open and pattern coding will be implemented to build a theory that identifies and connects relationships among other isolated e-textile circuit development theories. The suggested software for the coding process includes NVivo, MAXQDA, or Dedoose. The EDA framework is the expected result of this phase in the study.

3.1.3 Possible Limitation and Mitigation Strategy. A foreseeable limitation is that the data collected from other studies might be limited to the application it was collected for. The mitigation strategy

for such limitation is gathering or collating data from different studies, interviewing experts, and the researcher's experience might limit that effect.

3.2 Phase 2 – Experimental Design

This stage focuses on generating design rules that guide the development of e-textile circuits. A few variable relationships identified in the previous phase will be characterized in depth to develop design rules for one sector of the larger framework (likely routing and placement for stitched woven e-textiles, depending on the results from phase 1). In this second phase, we will characterize selected variables and variable relationships; variable relationships such as the influence of trace spacing on electrical resistance, stitch density on electrical resistance, pad geometry on solderability performance, stitch termination types on trace spacing, yarn type on trace spacing, etc. We will use quantitative methods at this phase to check for correlation, not cause-and-effect among variables. The decision of what variables to study cannot be made before the full EDA framework has been constructed. Here, some hypothetical examples are used to illustrate the kind of approach that will be taken. For example: if three variables are selected from the previous phase for circuit routing, say, for instance, trace spacing and thread types (independent variables) and conductivity (dependent variable); and for component placement, say, for instance, stitch pad

dimension (independent variable) and durability (dependent variable) are eventually selected as the variables to characterize their relationships and develop design rules in an e-textile circuit, then this is the approach I might take in going about the characterization process:

3.2.1 Example Characterization Study. How does trace spacing among different thread type influence conductivity in a stitched electrical circuit?

Procedure. For this example, we will first define a test circuit layout to measure the relationship between thread characteristics and trace spacing. In this case, we might assume a basic LED circuit layout. Afterward, we will select three thread types and stitch the circuit five times, each time with a different three trace width spacings in the circuit layout. That will make about 45 trials in total. Then we will test their conductivity. The conductivity parameters here will be to test for shorts and check their electrical resistances. This method was adopted from the [21] study. But in our case, we are improving on [21] to compare more than one thread type and compile insights into design rules. An encoded stereo microscope (Leica M165C) / a high-tech camera that can capture details would be used to observe the circuit images and measure the diameter of the stitched lines. A Fluke 8846A Digital Precision Multimeter would be used to measure the electrical resistance of the circuit.

Parameters of Evaluation. For our example case of trace spacing, we might evaluate the performance of the stitched circuit electrically, photographically, and statistically. An analysis of variance test (ANOVA) could be conducted to test the statistical significance ($\alpha = 0.05$) of the resistance values based on the trace spacing. The resistance values could be plotted against each stitched thread and stitch spacing. For the image analysis, the image of the stitched circuit could be interposed on the CAD circuit layout to allow measurement of the variation between the two images in pixels and mm.

3.2.2 Example Characterization Study. What stitch pad dimension size is needed to reduce breakage among 2-pin and 4-pin component packages?

Goal. This example aims to observe the association between stitch pads and the durability of component packages that can be useful for standardization. [21] investigated solder joints of components and breakage. We are improving this insight from [21] into a reliable and repeatable experimental process. The independent variables are the stitch pad dimension and component lead pin types. The dependent variable is durability (parameter - Force in N).

Procedure. For this example, first, we would define the dimensions of our pad for the circuit – (e.g., 2mmx2mm, 3mmx3mm, 4mm x4mm). The pads are usually stitched in a rectangular form. Although there are different pad shapes in regular PCB manufacturing, we would likely stick with the rectangular shape for our study to minimize variables. Then we would stitch different pads using one conductive thread type. The smallest pad, e.g., 2mm x 2mm, would be the smallest stitch pad the pattern stitching machine can stitch. Leads from Small Outline Integrated Circuit (SOIC) standard component packages would be soldered to these different pads, and

their durability would be evaluated by pulling the pad and textile in opposite directions using the Instron machine. We would perform multiple (three) runs of this test and record the force at which the solder joint breaks.

Data Analysis. Multi-linear regression analysis can be used to model these relationships, but before modeling, we will check for collinearity among our independent variables. Our evaluation parameter will be adjusted R-squared ($\alpha = 0.05$).

3.3 Phase 3 – Demo Application

3.3.1 Example Characterization Study. How well does the solar circuit developed by the above design guidelines work accurately and repeatably according to the previously characterized performance metrics after re-creating the circuit five times?

Goal. This phase involves the practical application of our model developed in Phase 2. The demo application was selected to build on our prior work of developing solar e-textile circuits and assess the robustness and generality of our design rules.

Measurement Methods. We will implement our design rules in a solar circuit by varying three stitch pads and three different solar cell packages using a Vectran thread. The stitch pads were characterized previously, while the solar cell packages have not been characterized. Here, we are checking for consistency of the characterized stitch pads for the new set of component packages. The solar circuit will be created five times using the same guideline in Phase 2, and the five stitched solar circuits will be placed in an Instron machine to test for durability and then test the circuit's functionality.

Parameters of Evaluation. We will use an Intraclass Correlation Coefficient (ICC) to assess the level of agreement or consistency of the five solar circuits developed. The ICC is a statistical measure used to assess the proportion of total variance attributed to true between sample variation as opposed to measurement readings errors. The ICC test varies between 0-1, where higher values indicate greater reliability of results taken. Additionally, we will validate our statistical results by measuring and comparing the efficiency of the solar textile circuits.

4 EXPECTED CONTRIBUTIONS

- **Developing an EDA framework for e-textiles** will help **standardize** procedures for routing and placement in e-textile circuit development.
- **Characterization of fabrication variables** and relationships will help streamline e-textile circuit design.
- Demonstration of improved **manufacturability** of e-textile circuits through design rules will advance the state of the art in e-textile manufacturing.

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